

Enhancing Multimedia Forensics and Authentication Analysis through Enhanced Explainability and Summarization of Results

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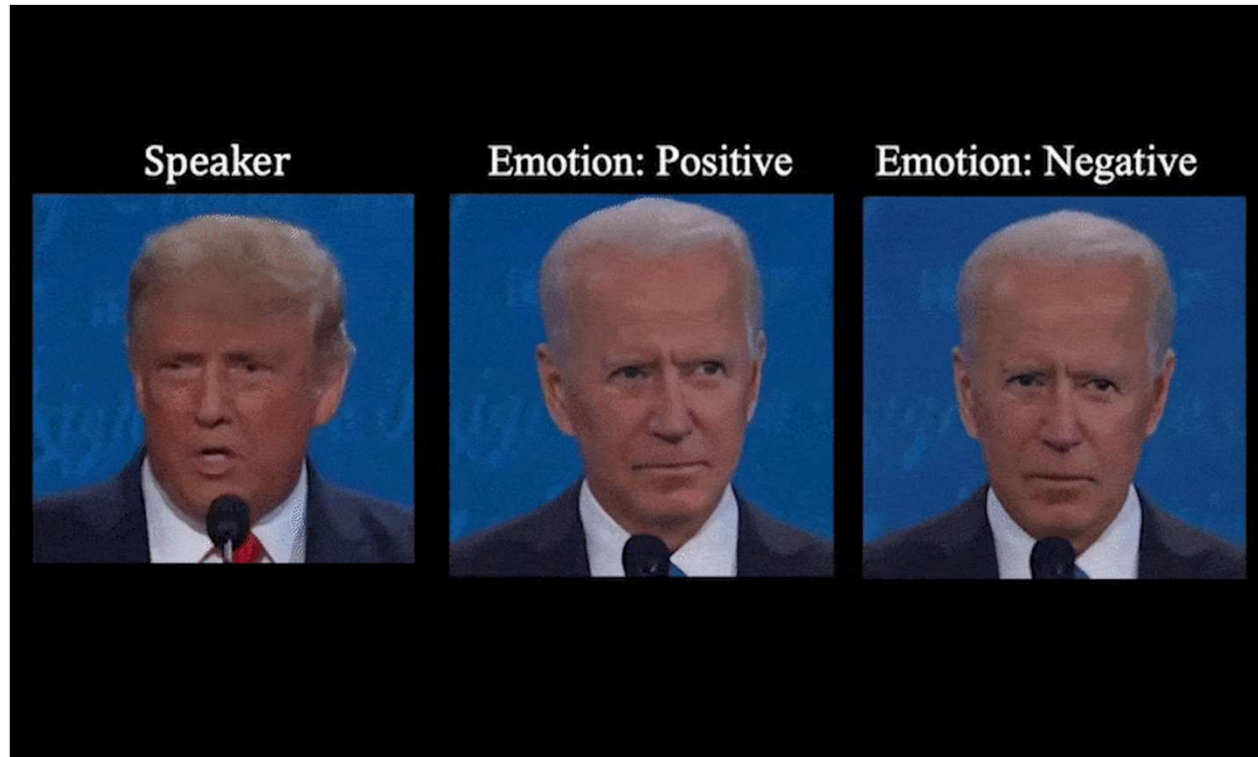
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Problem Mentors

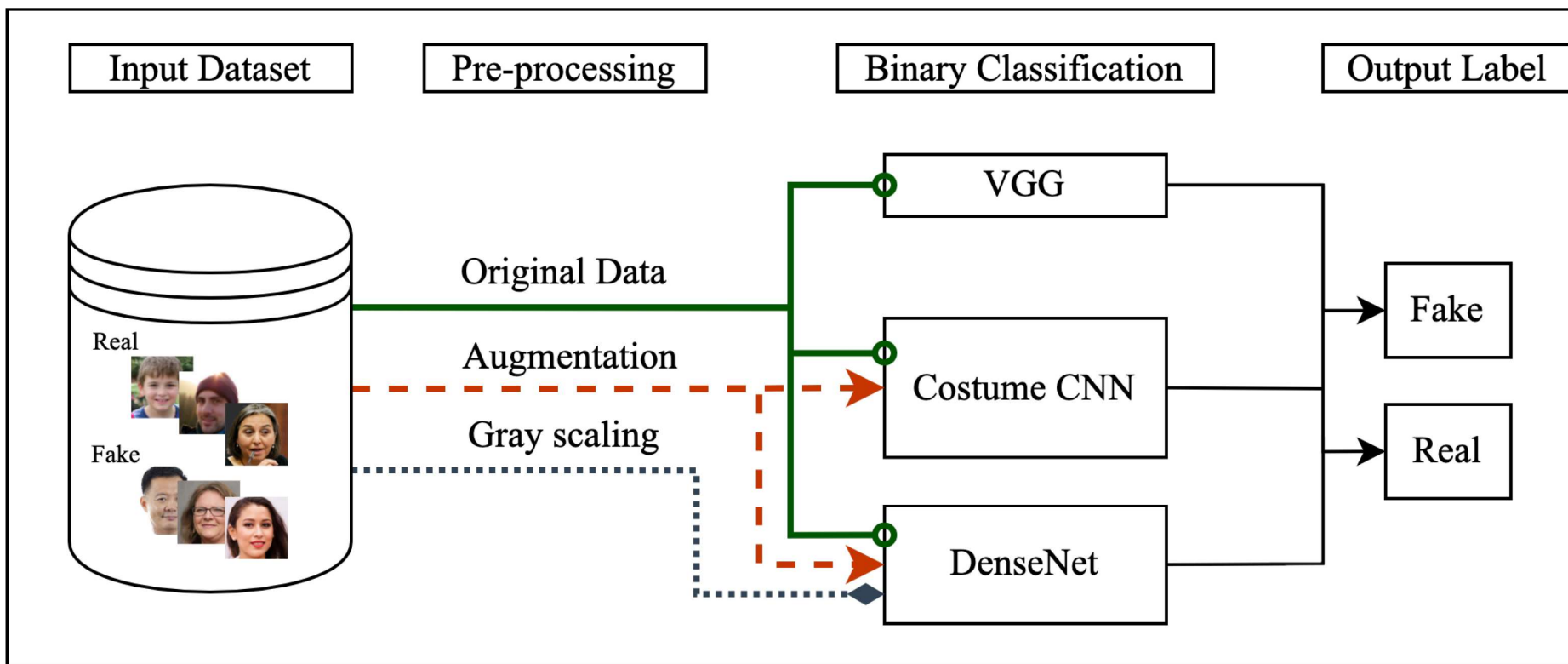
Part 1 Introduction of Explainable Forensics

Video Manipulation Detection is crucial



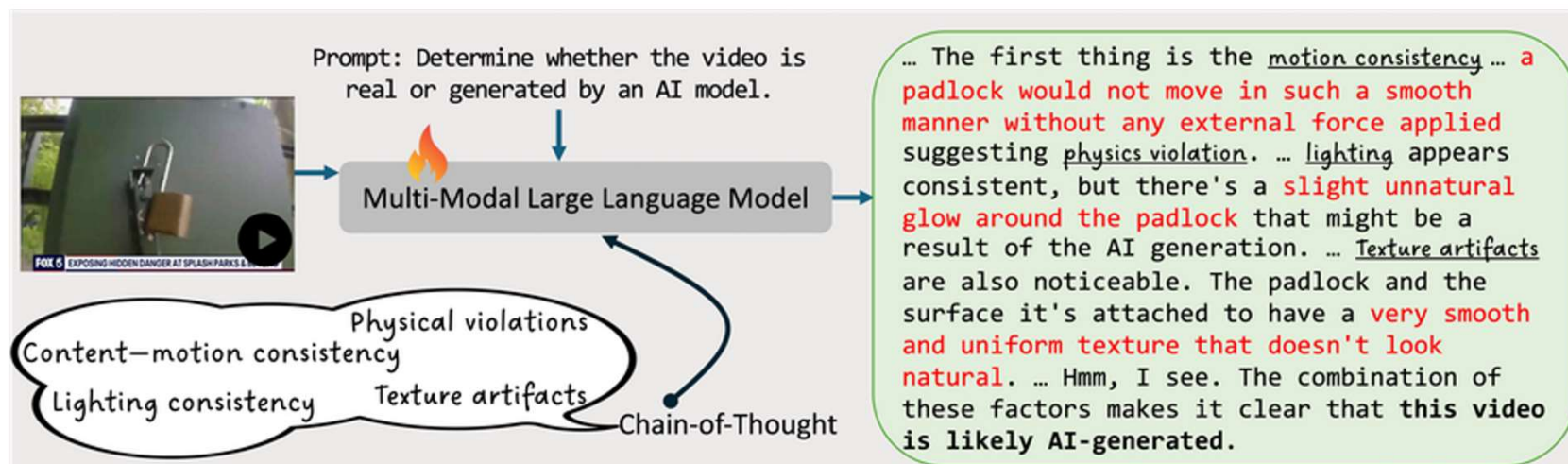
- Altered videos can be extremely dangerous

Current detection tools lack explanation



What is Explainable Forensics AI

- AI analysis for generating explanations



Part 2 Towards General-manipulation Forensics Detection

From face-manipulated to general-manipulated dataset

ExDDV



There are small left-right jitters of the eyes. The mouth shows distortions.



There is distortion of all facial features caused by overlapping artifacts. The eyebrows are darting up and down. A residual set of eyebrows is noticeable. Residuals from the source face flash across at times causing a change in the look of the features.



The mouth is blurred of a few moments. The details of the face are not sharp.



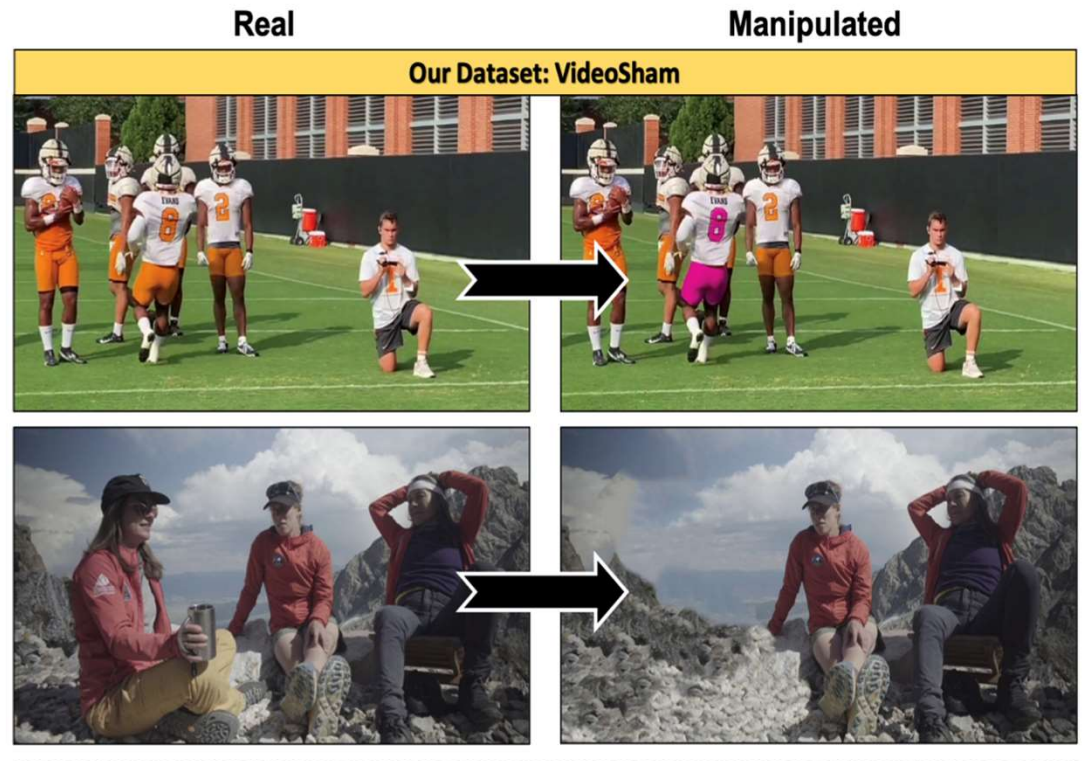
This video is visually coherent.

VideoSham (w. Explanation)



Selected Dataset: VideoSham

- Adobe Research Team
- Beyond Faces
- Total 6 manipulations



Manipulation I: Adding an entity/subject

Real



Manipulated



Description:

Around 0:42, there is a person in a black coat who passes the last turnstile, can you cut this person out, and make them cross the turnstile at a different time and different turnstile in the video

Manipulation 2: Removing an entity/subject

Real



Manipulated



Description:

Remove the man on the left side (00:00 - 00:05)

Manipulation 3: **Background/Color Change**

Real



Manipulated



Description:

Change the color of the cabinets behind the man

Manipulation 4: **Text Replaced/Added**

Real



*A conversation
about CRPS*

Manipulated



lorem ipsum

Description:

0:00 to 0:06 change the text in the blue box to lorem ipsum

Manipulation 5: Inter-Frame Manipulations

Real



Manipulated



Description:

Duplicate random frames in between to make it into a 10 second video (00:10 - 00:16)

Manipulation 6: **Audio Replaced/Added**

Real



Manipulated



Description:

Trim to (3:05 - 3:10) and remove the guitar noise. Place the noise of a cow moo-ing instead.

Annotation Motivation

Real



Manipulated



Description:

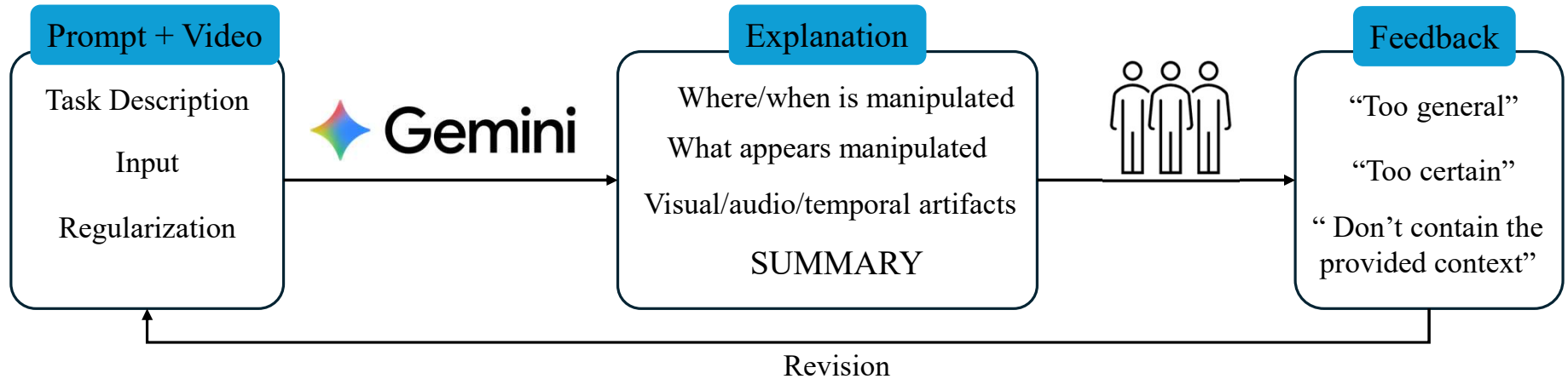
Trim to (3:05 - 3:10) and remove the guitar noise. Place the noise of a cow moo-ing instead.



Explanation:

The audio track has been replaced, introducing the sound of a cow mooing which is inconsistent with the visual action of a guitar being played.

Annotation Loop



Version 1: The primary manipulation is the removal of the man on the left. The most evident visual anomaly is the lack of interaction with the man on the right, which highlights the missing participant. Slight inconsistencies in texture, sharpness, or blurring may appear along the boundary of the area where the man was removed, but the rest of the scene appears unaffected.



Version N: The video shows signs of digital manipulation, with a subject on the left side of the frame likely removed, evidenced by spatial inconsistencies, blurring artifacts, and a missing interaction dynamic.

Final Prompt

➤ Task Description

You are a video forensics expert analyzing a single processed video for signs of digital manipulation. Provide a comprehensive forensic analysis based only on what the processed video shows:

➤ Input Description

Use the provided context and suspected manipulation ONLY to guide where you look; do not repeat or rely on them in your conclusions.

Context (do not cite directly): {task_description}

Suspected manipulation (do not cite directly): {manipulation_type}

Processed video: {<video></video>}

Unedited video: {<video></video>}

➤ Regularization

IMPORTANT: Base findings purely on evidence within this single video. Do not mention any reference/original footage. Phrase conclusions with uncertainty (e.g., "appears", "seems") rather than definitive claims ("has been"). If unsure, state that the evidence is unclear and avoid inventing objects (e.g., do not describe removed items that are not visible).

When describing entities or interactions, avoid assuming identities, genders, or roles unless they are explicitly evident. Use neutral descriptors like "a subject", "an individual", "a figure", "an object", or "something/someone" when certainty is low. If an element is implied but not visible, refer to it generically (e.g., "an unseen subject" or "an unseen object") instead of naming specific categories.

Input & Output Example

Prompt

...

Context (do not cite directly):
{Trim to (3:05 - 3:10) and remove the guitar noise. Place the noise of a cow moo-ing instead.}

Suspected manipulation (do not cite directly): {Audio replaced/Audio added}

Processed video: {<video></video>}

Unedited video: {<video></video>}

...

Processed Video



Unedited Video



WHAT APPEARS MANIPULATED:

The primary manipulation is auditory rather than visual. While the video content—showing a person playing a guitar—remains visually identical to the reference, the audio track has been completely substituted. The original instrumental music has been removed and replaced with unrelated sound effects.

TEMPORAL ARTIFACTS:

xxx

AUDIO-VISUAL SYNC:

xxx

FORENSIC INDICATORS:

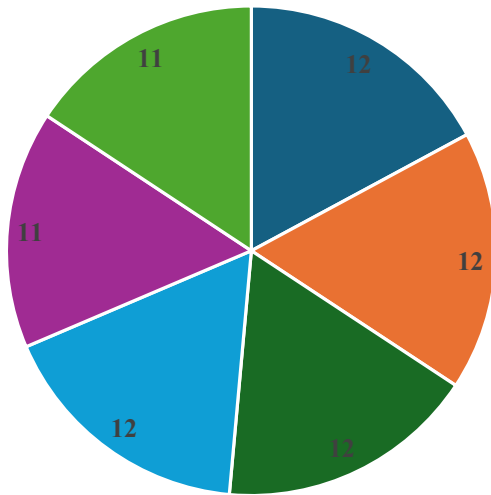
- xxx

SUMMARY:

The suspect video has undergone audio manipulation where the original guitar track was replaced with unrelated animal vocalizations, resulting in a complete disconnect between the visual actions and the accompanying sound.

Dataset Statistics

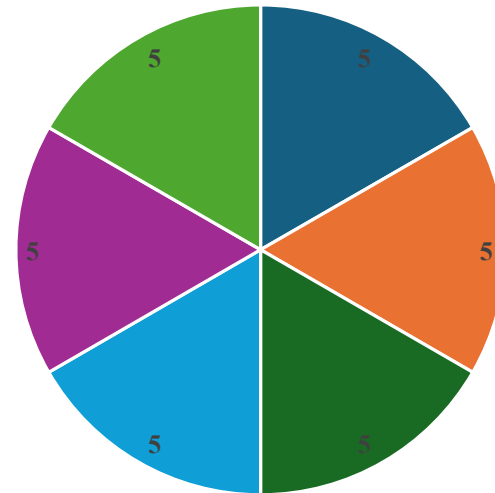
Training Set # = 70



- Removing an entity
- Adding an entity
- Audio replaced/Audio added
- Frame Manipulation
- Background/Color Change
- Text Replaced/Text Added

Average Explanation Length: 32.9 words

Testing Set # = 30



- Removing an entity
- Adding an entity
- Audio replaced/Audio added
- Frame Manipulation
- Background/Color Change
- Text Replaced/Text Added

Average Explanation Length: 32.5 words

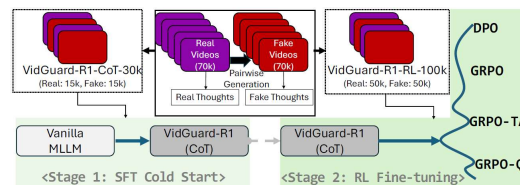
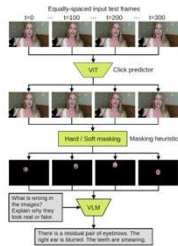
Part 3 General Manipulation Detection Model

Experiment Settings

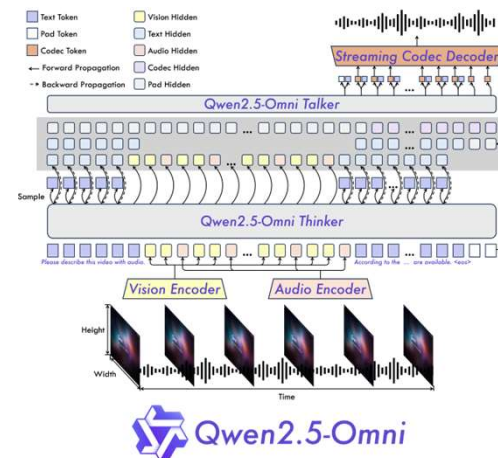
➤ Models

Models in prior works don't support omni-modality input (i.e. video and audio input)

ExDDV:



VidGuard-R1



➤ Metrics

BLEU1: how many words in the prediction overlap the ground truth

Semantic Cosine: cosine similarity between sentence embeddings of prediction and reference; higher means closer meaning even if wording differs.

Accuracy: whether the predicted explanation is consistent with the ground truth judged by human.

Zero-Shot Model Evaluation

Prompt

You are a video forensics expert analyzing a single video for signs of digital manipulation. If manipulation is suspected, provide one sentence forensic analysis based on this video, including: WHERE/WHEN: provide coordinates or time ranges of anomalies. WHAT APPEARS MANIPULATED: Describe suspected inconsistencies with cautious language.

Processed Video



Qwen2.5-Omni (7B)

Prediction

The video appears to have been manipulated around the 1.5-second mark, where the background lighting changes abruptly, suggesting a possible overlay or digital alteration.

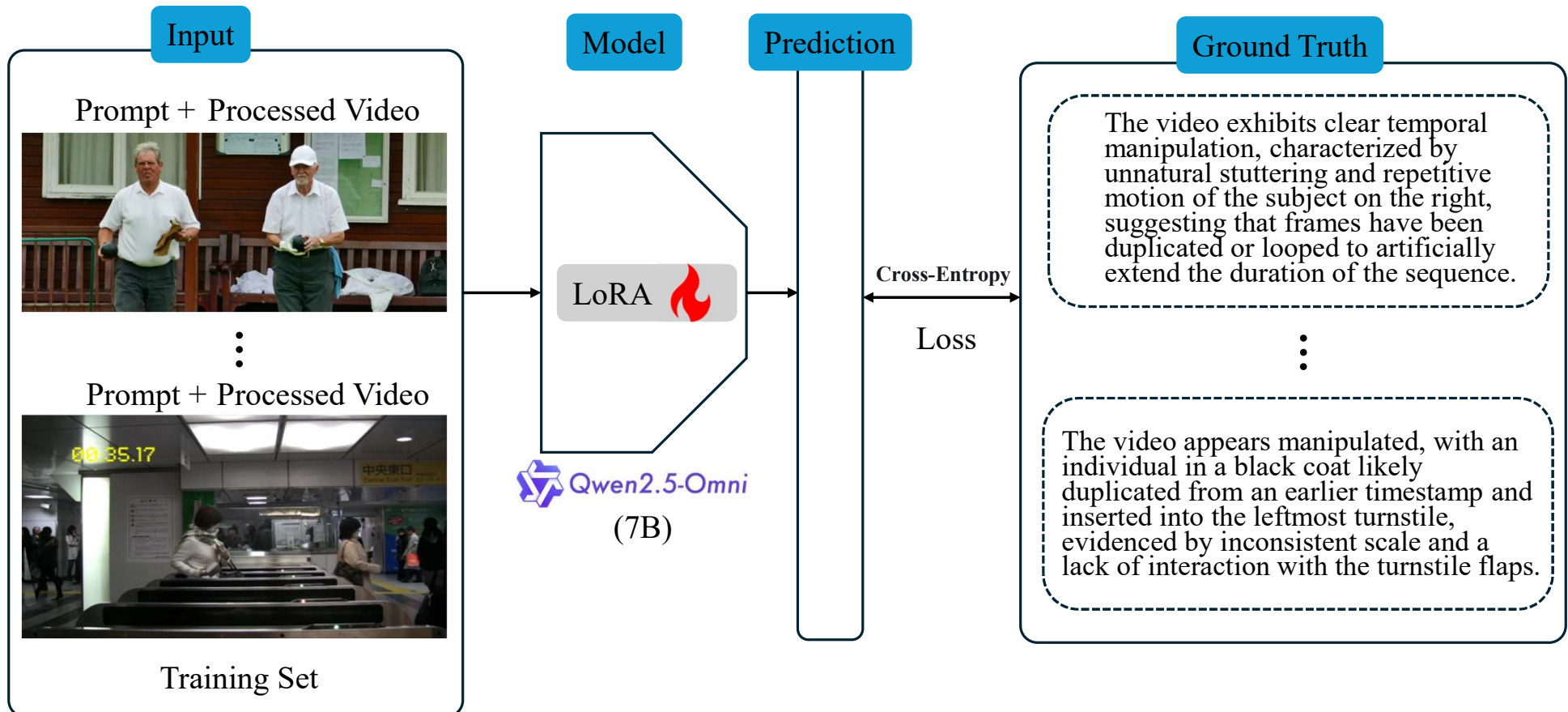
Ground Truth

The video appears manipulated, as an individual on the right side of the frame is missing in the final segment, causing the remaining subject to interact with an empty space while the removed individual's audio remains.

BLEU1	Semantic Cosine	Accuracy
0.2326	0.5766	16.67%

Note that the human accuracy is 83.33%.

Omni-Model Finetuning Pipeline



Training

➤ Main Configuration

Learning rate: $1e-4$

Optimizer: AdamW

Epochs: 3

Batch size: 1

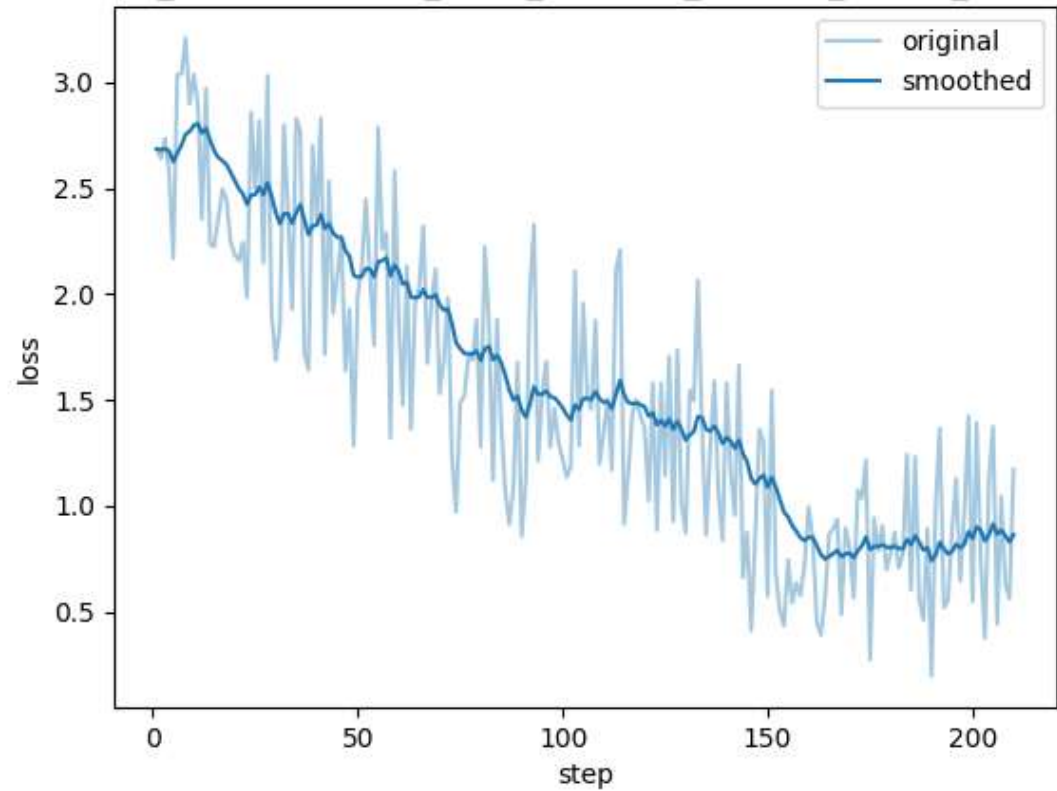
➤ More Configurations

Warmup ratio: 0.1

Scheduler: cosine

Vision & Audio Tower: freeze

Video FPS: 2



Finetuned Model Evaluation

Prompt

You are a video forensics expert analyzing a single video for signs of digital manipulation. If manipulation is suspected, provide one sentence forensic analysis based on this video, including: WHERE/WHEN: provide coordinates or time ranges of anomalies. WHAT APPEARS MANIPULATED: Describe suspected inconsistencies with cautious language.

Processed Video



Qwen2.5-Omni (7B) Finetuned

Prediction

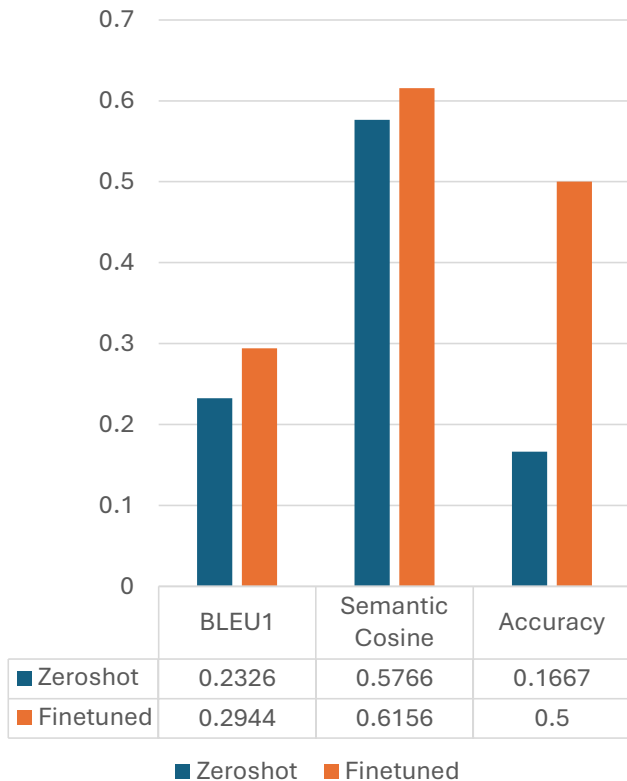
The video appears manipulated, as the initial footage of a natural stream is abruptly replaced by a looping ocean waves background, evidenced by the complete lack of water flow in the latter segment.

Ground Truth

The video appears to undergo an audio manipulation at 00:21, where the original ambient sound and dialogue are abruptly replaced by the incongruent sound of crashing waves, resulting in a complete desynchronization with the visual environment and the speaking subjects lip movements.

BLEU1	Semantic Cosine	Accuracy
0.2944	0.6156	50%

Zero-Shot & Finetuned Comparison



Processed Video



Ground Truth

The video appears manipulated, as the on-screen text identifying the subject changes abruptly from Sue Grogan to Juan Rowling, creating a clear contradiction with the spoken dialogue where the individual identifies herself as Sue Grogan.

Zero-Shot

The background appears to be digitally manipulated, as the lighting and shadows on the foliage do not align with the natural lighting conditions visible in the rest of the video.

Finetuned

The video appears manipulated, as the text overlay identifying the speaker as Sue Grogan from Joined Up Working Associates is duplicated and applied to an individual with a completely different name and title, indicating a case of identity substitution.

Failure Case



Ground Truth

The footage displays persistent jerky motion and temporal discontinuities throughout, strongly suggesting that frames have been systematically dropped, resulting in an unnatural, accelerated playback speed.

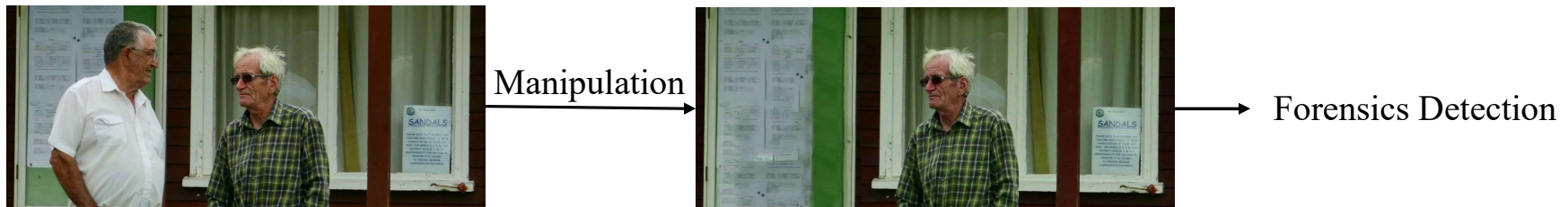
Prediction

The video appears manipulated, as the distant buildings visible in the second segment **seem to have been digitally composited into the background**, evidenced by their lack of interaction with the water and the subject failure to react to their presence.

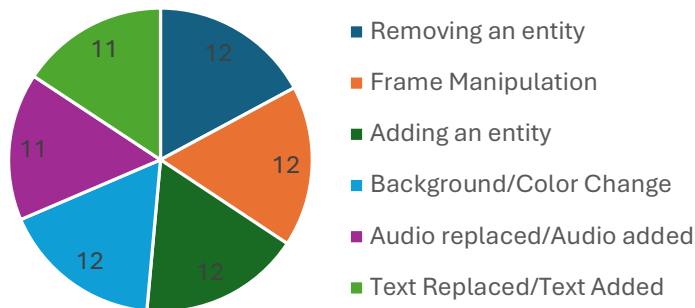
- **Frame manipulation is difficult for AI assistant to detect while easy for human.**

Summary

- 1. We introduced a research task towards general-manipulation forensics detection.

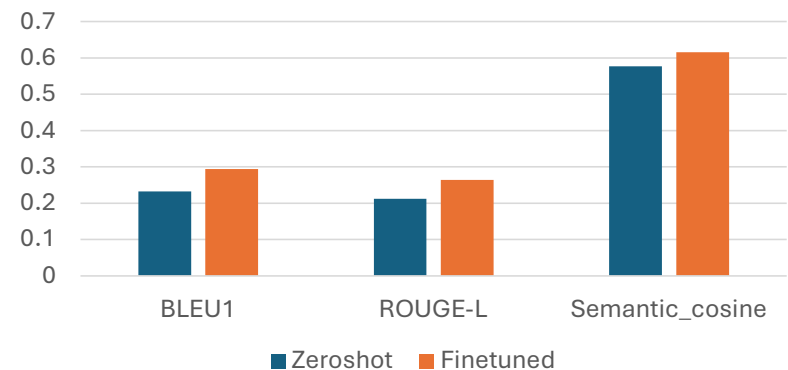


- 2. We proposed a data generation pipeline based on Gemini and curated a dataset for general-manipulation forensics.



Average Explanation Length: 32 words

- 3. We evaluated open-source omni-modality model (i.e. Qwen-2.5-Omni-7B) on the curated dataset. Experiments show finetuned model achieves better performance than the non-finetuned one.



Thank you!

Questions are welcome!